Assignment 3 (Himanshu Sharma - 230474)

Ans 1:

$E_1(t) = E_1 \cos(\omega_0 t) \rightarrow \tau_1$

$E_2(t) = E_2 \cos(\omega_0 t + \phi) \rightarrow \tau_2$

$$\Omega = \frac{\mu_{12} E_0}{\hbar}, \quad \theta = \int_0^{\tau} \Omega \, dt = \frac{\mu_{12} E_0 \tau}{\hbar}$$

$$|g\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad |e\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$U(\theta, \phi) = \exp\left[i \frac{\theta}{2} (\sigma_x \cos\phi + \sigma_y \sin\phi)\right]$$

$$= \begin{bmatrix} \cos(\theta/2) & i e^{-i\phi} \sin(\theta/2) \\ i e^{i\phi} \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

$$\frac{\mu_{12}}{\hbar} = \frac{2 \times 10^{-29}}{1.054 \times 10^{-34}} \approx 1.8965 \times 10^{5} \text{ rad}/(\text{V/ms})$$

(a) $\theta_1 = \frac{\mu_{12} E_1 \tau_1}{\hbar} = 0.785 \text{ rad} \approx \pi/4$

$\theta_2 = \frac{\mu_{12} E_2 \tau_2}{\hbar} = 2.355 \text{ rad} \approx 3\pi/4$

Since $\phi = 0$, E_1 and E_2 are in same phase. Hence they correspond to rotations around the same axis (x-axis)

$\theta_{total} = \theta_1 + \theta_2 = \pi$

$$|\psi_{fin}\rangle = U(\pi, 0)|g\rangle = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} = \boxed{i|e\rangle}$$

(b) $\theta_1 = \dfrac{\mu_{12} E_1 \tau_1}{\hbar} = 1.57 \text{ rad} \approx \dfrac{\pi}{2} \text{ rad}$

$\theta_2 = \dfrac{\mu_{12} E_2 \tau_2}{\hbar} = 3.14 \text{ rad} \approx \pi \text{ rad}$

First pulse is $\pi/2$ rad rot. around x-axis

$|\psi_1\rangle = U(\pi/2, 0)|g\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$

$|\psi_2\rangle = U(\pi, \pi/2)|\psi_1\rangle = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ i \end{pmatrix}\frac{1}{\sqrt{2}} = \frac{i}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$

$\boxed{|\psi_{fin}\rangle = \frac{1}{\sqrt{2}}(|g\rangle + i|e\rangle)}$

(c) $\theta_1 = \dfrac{\mu_{12} E_1 \tau_1}{\hbar} = 1.57 \text{ rad} \approx \pi/2 \text{ rad}$

$\theta_2 = \dfrac{\mu_{12} E_2 \tau_2}{\hbar} \approx 3.14 \text{ rad} \approx \pi \text{ rad}$

$|\psi_1\rangle = 1/\sqrt{2}\begin{pmatrix} 1 \\ i \end{pmatrix}$ just like case (b)

$|\psi_{fin}\rangle = U(\pi, \pi/4)|\psi_1\rangle = \begin{pmatrix} 0 & e^{i\pi/4} \\ e^{3i\pi/4} & 0 \end{pmatrix}\begin{pmatrix} 1 \\ i \end{pmatrix}\frac{1}{\sqrt{2}}$

$\Rightarrow |\psi_{fin}\rangle = \dfrac{e^{3i\pi/4}}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Physical state : $\boxed{\frac{1}{\sqrt{2}}(|g\rangle + |e\rangle)}$

Ans 2 :

$$\frac{d}{dt}\begin{pmatrix} u \\ v \\ w \end{pmatrix} = -\begin{pmatrix} \Omega_0 \\ 0 \\ \delta \end{pmatrix} \times \begin{pmatrix} u \\ v \\ w \end{pmatrix} - \gamma \begin{pmatrix} u/2 \\ v/2 \\ w+1 \end{pmatrix}$$

(a) $$\frac{d}{dt}\begin{pmatrix} u \\ v \\ w \end{pmatrix} = -\gamma \begin{pmatrix} u/2 \\ v/2 \\ w+1 \end{pmatrix}$$

$\Rightarrow$ $u(t) = u(0)\exp(-\gamma t/2)$

$v(t) = v(0)\exp(-\gamma t/2)$

$w(t) = (w(0)+1)\exp(-\gamma t) - 1$

(b) excited state $= (0\ 0\ 1)^T$ $(t=0)$

$\Rightarrow$ $u(t) = 0$, $v(t) = 0$ [plots at the end]

$w(t) = 2\exp(-\gamma t) - 1$

at $t = \ln 2/\gamma$: $2^{(1-\gamma/\gamma)} - 1 = \boxed{w(t) = 0}$

at $t \to \infty$: $\boxed{w(t) = -1}$

(c) $R(0) = (1\ 0\ 0)^T$

$u(t) = \exp(-\gamma t/2)$ $v(t) = 0$

$w(t) = \exp(-\gamma t) - 1$ [plots at the end]

$R(\ln 2/\gamma) = (1/\sqrt{2}\ \ 0\ \ -1/2)^T$

$R(\infty) = (0\ \ 0\ \ -1)^T$

(d) when $\Omega_0 > 0$:

$$-\begin{pmatrix} \Omega_0 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} u \\ v \\ w \end{pmatrix} = -\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \Omega_0 & 0 & 0 \\ u & v & w \end{vmatrix} = \begin{pmatrix} +0 \\ +\Omega_0 w \\ -\Omega_0 v \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{pmatrix} = \begin{pmatrix} -u\gamma/2 \\ \Omega_0 w - \gamma v/2 \\ -\gamma(w+1) - \Omega_0 v \end{pmatrix} \Rightarrow u(t) = u(0)\, e^{-\gamma t/2}$$

$\cancel{\Rightarrow u(t\to\infty) = u(0)}$

in steady state: $\dot{u} = \dot{v} = \dot{w} = 0$

$\Rightarrow \boxed{u_{ss}(t) = 0}$

$\Omega_0 w_{ss}(t) - \gamma v_{ss}/2 = 0 \Rightarrow \boxed{w_{ss} = \dfrac{\gamma v_{ss}}{2\Omega_0}}$

$-\gamma w_{ss} - \gamma - \Omega_0 v_{ss} = 0$

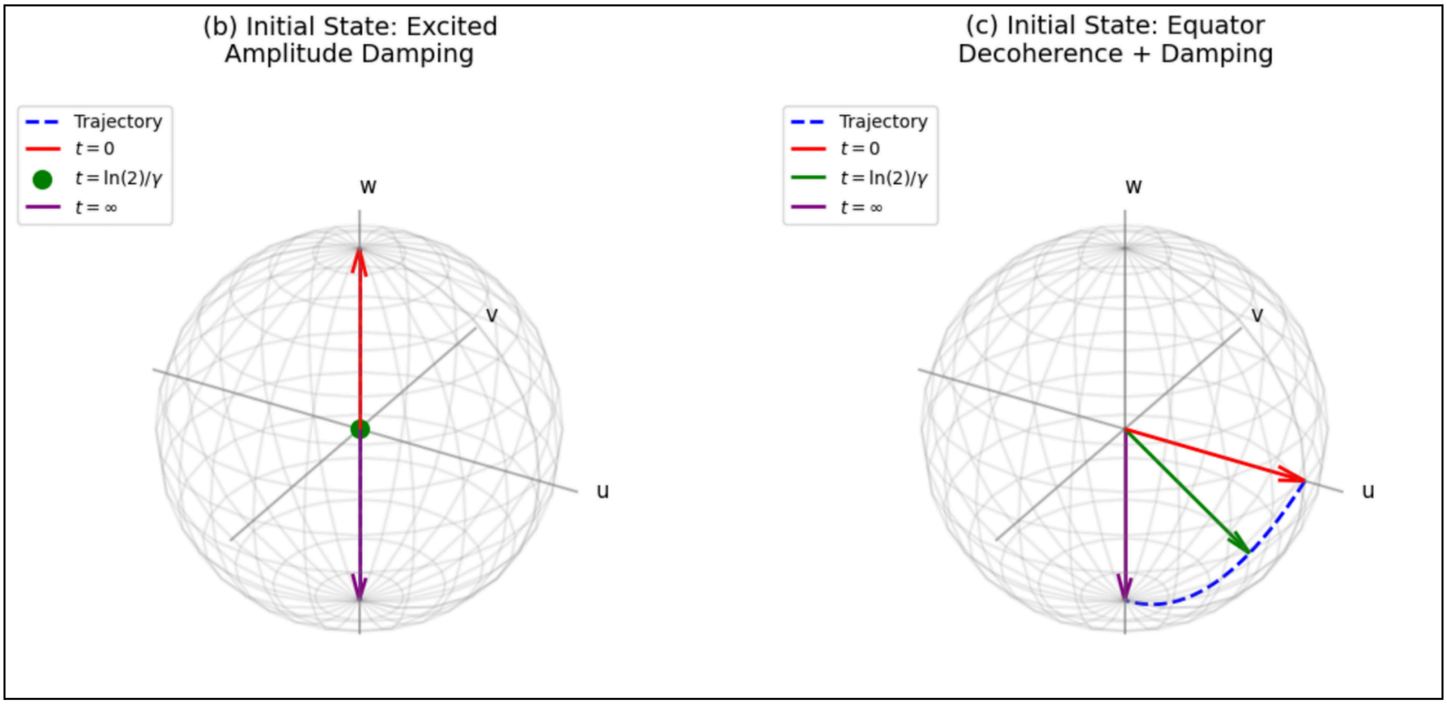
$\Rightarrow \dfrac{-\gamma^2 v_{ss}}{2\Omega_0} - \gamma - \Omega_0 v_{ss} = 0$

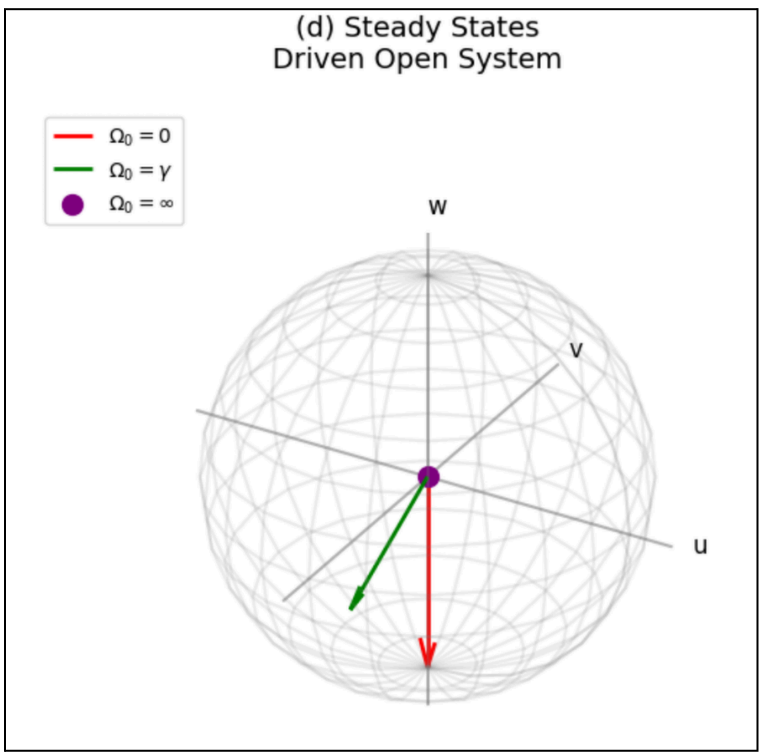
$\Rightarrow \boxed{v_{ss} = \dfrac{-2\Omega_0\gamma}{2\Omega_0^2 + \gamma^2}}$ $\boxed{w_{ss} = \dfrac{-\gamma^2}{2\Omega_0^2 + \gamma^2}}$

(i) $R_{ss} = (0 \;\; 0 \;\; -1)^T$

(ii) $R_{ss} = (0 \;\; -2/3 \;\; -1/3)^T$ [plots at the end]

(iii) $R_{ss} = (0 \;\; 0 \;\; 0)^T$


(b) Initial State: Excited
Amplitude Damping
Trajectory
$t = 0$
$t = \ln(2)/\gamma$
$t = \infty$
w
v
u
(c) Initial State: Equator
Decoherence + Damping
Trajectory
$t = 0$
$t = \ln(2)/\gamma$
$t = \infty$
w
v
u


(d) Steady States
Driven Open System
$\Omega_0 = 0$
$\Omega_0 = \gamma$
$\Omega_0 = \infty$
w
v
u

Ans 3 : $E_a = \hbar\omega_a$, $E_b = \hbar\omega_b$ $E_c = \hbar\omega_c$

$$\omega_a - \omega_b = \omega_b - \omega_c = \nu$$

$$H_{int} = -d\cdot E(t) = -d\cdot E_0 \cos(\nu t)$$

$$H_0 = \hbar\omega_a |a\rangle\langle a| + \hbar\omega_b |b\rangle\langle b| + \hbar\omega_c |c\rangle\langle c|$$

$\langle a|d|b\rangle = d_{ab}$, $\langle b|d|c\rangle = d_{bc}$

$\langle a|d|c\rangle = 0$, $\langle i|d|i\rangle = 0$

$\Omega_{ab} = \dfrac{-d_{ab}E_0}{\hbar}$, $\Omega_{bc} = -\dfrac{d_{bc}E_0}{\hbar}$

$$|\psi(t)\rangle = c_a(t)e^{-i\omega_a t}|a\rangle + c_b(t)e^{-i\omega_b t}|b\rangle + c_c(t)e^{-i\omega_c t}|c\rangle$$

$$\frac{\partial}{\partial t}|\psi(t)\rangle = \frac{-i}{\hbar}(H_0 + H_I(t))|\psi(t)\rangle$$

substituting $|\psi(t)\rangle$ in TDSE, we obtain:

$$i\hbar\left(\dot{c}_a e^{-i\omega_a t}|a\rangle + \dot{c}_b e^{-i\omega_b t}|b\rangle + \dot{c}_c e^{-i\omega_c t}|c\rangle\right) = H_I(t)|\psi(t)\rangle$$

projecting the above individually into each basis:

$$i\hbar\,\dot{c}_a e^{-i\omega_a t} = c_b e^{-i\omega_b t}\langle a|H_I(t)|b\rangle$$

$$\Rightarrow i\dot{c}_a(t) = c_b(t)\exp(-i(\omega_b - \omega_a)t)\,\Omega_{ab}\cos(\nu t)$$

$\omega_a - \omega_b = \nu \Rightarrow$ resonance

$$\cos\nu t = (e^{i\nu t} + e^{-i\nu t})/2$$

$$\Rightarrow \boxed{i\dot{c}_a = \frac{\Omega_{ab}}{2}c_b(t)\{e^{2i\nu t} + 1\}} \quad (1)$$

similarly, for $|b\rangle$:

$$\boxed{i\dot{C}_b(t) = \frac{\Omega_{ab}}{2} C_a(t)\left[1+e^{-2i\nu t}\right] + \frac{\Omega_{bc}}{2} C_c(t)\left[e^{2i\nu t}+1\right]} \quad \hookrightarrow ②$$

and for $|c\rangle$:

$$\boxed{i\dot{C}_c(t) = \frac{\Omega_{bc}}{2} C_b(t)\left\{1+e^{-2i\nu t}\right\}} \quad — ③$$

Now applying RWA to 1, 2, 3:

$$i\dot{C}_a(t) = \frac{\Omega_{ab} C_b(t)}{2} \qquad — 4a$$

$$i\dot{C}_b(t) = \frac{\Omega_{ab} C_a(t) + \Omega_{bc} C_c(t)}{2} \qquad — 4b$$

$$i\dot{C}_c(t) = \frac{\Omega_{ab} C_b(t)}{2} \qquad — 4c$$

The effective, time independent Hamiltonian in the interaction picture is: $i\hbar \dot{C}(t) = H_{RWC}\, C(t)$

$$\Rightarrow H_{RWA} = \frac{\hbar}{2}\begin{bmatrix} 0 & \Omega_{ab} & 0 \\ \Omega_{ab} & 0 & \Omega_{bc} \\ 0 & \Omega_{bc} & 0 \end{bmatrix}$$

now, let $\Omega_{ab} = \Omega_{bc} \equiv \Omega$, $W = \Omega/2$

$$\Rightarrow \dot{C}_a = -iW C_b, \quad \dot{C}_b = -iW(C_a + C_c), \quad \dot{C}_c = -iW C_b$$

note: $\dot{C}_a = \dot{C}_c \Rightarrow \ddot{C}_b = -iW(-2iW C_b)$

$$\Rightarrow \ddot{C}_b = -2W^2 C_b \Rightarrow \boxed{C_b(t) = A\cos(\sqrt{2}\,Wt) + B\sin(\sqrt{2}\,Wt)}$$

initial conditions : $c_a(0) = 0$, $c_b(0) = 0$, $c_c(0) = 1$

$$\Rightarrow \quad c_b(t) = B\sin(\sqrt{2}\,Wt)$$

also $\quad \dot{c}_b(t) = B\sqrt{2}\,W\cos(\sqrt{2}\,Wt) = -iW(c_a + c_c)$

$$\cancel{\ddot{c}_b(t) = -2W^2 B\sin(\sqrt{2}\,Wt) = -2W^2}$$

now at $t=0$: $\quad \dot{c}_b(0) = \sqrt{2}BW \cancel{\cos(\sqrt{2}W)} = -iW$

$$\Rightarrow \boxed{B = -i/\sqrt{2}}$$

$$\boxed{c_b(t) = \frac{-i}{\sqrt{2}}\sin(\sqrt{2}\,Wt)}$$

$$\boxed{c_a(t) = \frac{1}{2}\cos(\sqrt{2}\,Wt) - \frac{1}{2} = -\sin^2\left(\frac{Wt}{\sqrt{2}}\right)}$$

$$\boxed{c_c(t) = \frac{1}{2}\cos(\sqrt{2}\,Wt) + \frac{1}{2} = \cos^2\left(\frac{Wt}{\sqrt{2}}\right)}$$

☆ $P_c(t) = |c_c(t)|^2 = \cos^4(\Omega t/2\sqrt{2})$

☆ $P_a(t) = |c_a(t)|^2 = \sin^4(\Omega t/2\sqrt{2})$

☆ $P_b(t) = |c_b(t)|^2 = \sin^2(t\Omega/\sqrt{2})/2$